

Digital Electronics (II SEM B.Tech. CSE)

Couse Code: NECE102



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Digital Electronics: Course Goals and Key Takeaways



Course Objective

To develop an understanding of digital electronics and its application which will be further useful to in the advanced courses of Electronics and Industrial Engineering.

Learning Outcomes

Upon successful completion of this course, students will:

- Gain information about the basic knowledge of digital electronics to design various types of digital circuits and systems.
- Learn how to analyse the performance and working of digital circuits.
- Develop their knowledge of digital electronics for industrial and robotic applications.

Digital Electronics: Key Topics at a Glance



Unit No.	Topics to be Covered	Lecture Hours	Learning Outcome
1.	Number System, Logic Gates, Boolean Algebra, NAND and NOR Implementation of Logic Functions, Determination of Boolean Functions from Digital Circuits.	08	Acquire an understanding of number system, basic logic gates and their applications to develop digital circuits and systems.
2.	Standard Representations of Logical Functions, SOP and POS form, Simplification of Logical Functions using K-map.	08	Acquire the knowledge to represent the logic functions in standard form and its minimization.
3.	Combinational Logic Circuit: Adder, Subtractor, Magnitude Comparator, Code Converters etc. MSI Circuits: Multiplexers, Demultiplexers, Decoders, Encoders etc.	08	Understand the design and applications of combinational logic circuits.
4.	Sequential Circuits: Latches and Flip-flops, Counters, Shift Registers.	10	Acquire the ability to design and perform the analysis of different sequential circuits.
5.	Analysis of Sequential circuits: State tables and state diagrams, Timing Circuits	04	Acquire an understanding on the analysis of sequential circuits.
6.	Introduction to logic families, semiconductor memories, ROM, PAL, PLA and Microprocessor	04	Develop knowledge about the various techniques for IC designing for digital circuits.

Textbook:	Reference Books:
1. Floyd T.L., Digital Fundamentals, 10/2, Pearson Education, 2011.	1. Digital Logic and Computer Design by M. Morris Mano, Prentice –Hall, 2015. 2. Fundamental of Digital Electronics by A. Anand Kumar, 4/e Prentice Hall, 2016.

Digital Electronics: Lecture Plan



Sl. No.	Topics	No of classes (tentative)
1.	Introduction to Number Systems and Conversion Techniques	8
2.	Continue.....	
3.	1's and 2's complement	
4.	Logic Gates (AND, OR, NOT, NAND, NOR, XOR, XNOR)	
5.	Boolean Algebra: Laws and Theorems	
6.	Boolean Function Implementation Using NAND/NOR Gates	
7.	Determination of Boolean Functions from Digital Circuits	
8.	Continue.....	
9.	SOP Standard Forms of Logical Functions	8
10.	POS Standard Forms of Logical Functions	
11.	Simplification of Logical Functions Using Boolean Algebra	
12.	Introduction to Karnaugh Maps (K-Maps)	
13.	K-Map Simplification (2-variable Functions)	
14.	K-Map Simplification (3-variable Functions)	
15.	K-Map Simplification (4-variable Functions)	
16.	Don't Care Conditions in K-Map Simplification	

Digital Electronics: Lecture Plan



17.	Introduction to Combinational Logic Circuits	8
18.	Design of Adders (Half Adder, Full Adder)	
19.	Design of Subtractors (Half Subtractor, Full Subtractor)	
20.	Magnitude Comparator Design	
21.	Code Converters (Binary to Gray, Gray to Binary, BCD to Excess-3)	
22.	Multiplexers: Design and Applications	
23.	Demultiplexers: Design and Applications	
24.	Decoders and Encoders	
25.	Introduction to Sequential Circuits	10
26.	Latches: SR Latch, D Latch	
27.	Flip-Flops: SR, JK Flip-Flops	
28.	Flip-Flops: D and T Flip-Flops	
29.	Asynchronous Counters: Design and Operation	
30.	Continue.....	
31.	Synchronous Counters: Design and Operation	
32.	Continue.....	
33.	Shift Registers: Types and Applications	
34.	Continue.....	

Digital Electronics: Weightage



Weightage of Different Components

1.	Quiz/ Assignment	10 Marks
2.	Mid Semester Examination	30 Marks
3.	Quiz/ Assignment	10 Marks
4.	End Semester Examination	50 Marks
Total Marks = 100		

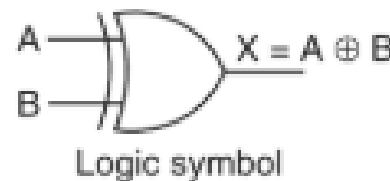
Logic Gates



Exclusive –OR (X-OR) GATE

- X-OR gate is a 2 input, one output logic circuit.
- Output assumes a logic 1 state when one and only one of its two inputs assumes a logic 1 state.
- When both the inputs assumes the logic 0 state, or when both inputs assume the logic 1 state, the output assumes a logic 0 state.
- Since an X-OR gate produces an output 1 only when the inputs are not equal, it is called an Anti-coincidence gate or inequality detector.
- The output of an X-OR gate is the modulo sum of its two inputs.
- The name Exclusive-OR is derived from the fact that its output is a 1, only when exclusively one of its inputs is a 1 (excludes the condition when both inputs are 1).

$$X = A \oplus B = \bar{A}B + A\bar{B}$$



Logic symbol

(a)

Inputs		Output
A	B	$X = A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

(b) Truth table



(c) X-OR gate as an inverter

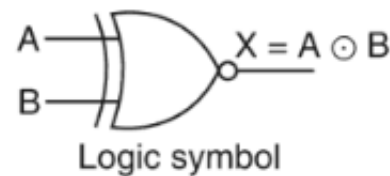
Logic Gates



Exclusive – NOR (X-NOR) GATE

- X-NOR gate is a 2 input, one output logic circuit.
- Output assumes a logic 1 state only when both inputs assume a zero state or when both the inputs assume a 1 state.
- Equality detector because it outputs a 1 only when its inputs are equal.

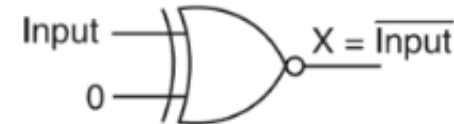
$$X = A \odot B = AB + \bar{A}\bar{B} = \overline{A \oplus B} = \overline{A\bar{B} + \bar{A}B}$$



(a)

Inputs		Output
A	B	$X = A \odot B$
0	0	1
0	1	0
1	0	0
1	1	1

(b) Truth table



(c) X-NOR gate as an inverter

Boolean Algebra



- It is the set of rules used to simplify the given logical expression without changing its functionality.
- $F = \overline{AB} + BC + ABC$
- Used when number of the variables are less

Axioms and Laws of Boolean Algebra

- *Axioms* or *postulates* of Boolean algebra are a set of logical expressions that we accept without proof and upon which we can build a set of useful theorems.
- Axioms are nothing more than the definitions of the three basic logic operations that we have already discussed: AND, OR, and INVERT. Each axiom can be interpreted as the outcome of an operation performed by a logic gate.

AND operation

Axiom 1: $0 \cdot 0 = 0$

Axiom 2: $0 \cdot 1 = 0$

Axiom 3: $1 \cdot 0 = 0$

Axiom 4: $1 \cdot 1 = 1$

OR operation

Axiom 5: $0 + 0 = 0$

Axiom 6: $0 + 1 = 1$

Axiom 7: $1 + 0 = 1$

Axiom 8: $1 + 1 = 1$

NOT operation

Axiom 9: $\overline{1} = 0$

Axiom 10: $\overline{0} = 1$

Boolean Algebra



Complementation Law

The term *complement* simply means to invert, i.e. to change 0s to 1s and 1s to 0s. The five laws of complementation are as follows:

Law 1: $\overline{0} = 1$

Law 2: $\overline{1} = 0$

Law 3: If $A = 0$, then $\overline{A} = 1$

Law 4: If $A = 1$, then $\overline{A} = 0$

Law 5: $\overline{\overline{A}} = A$ (double complementation law)

Notice that the double complementation does not change the function.

AND Laws

The four AND laws are as follows:

Law 1: $A \cdot 0 = 0$ (Null law)

Law 2: $A \cdot 1 = A$ (Identity law)

Law 3: $A \cdot A = A$

Law 4: $A \cdot \overline{A} = 0$

OR Laws

The four OR laws are as follows:

Law 1: $A + 0 = A$ (Null law)

Law 2: $A + 1 = 1$ (Identity law)

Law 3: $A + A = A$

Law 4: $A + \bar{A} = 1$

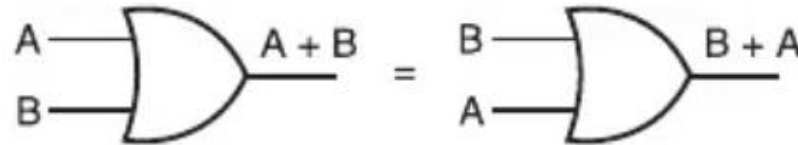
Boolean Algebra



Commutative Laws

Commutative laws allow change in position of AND or OR variables. There are two commutative laws.

Law 1: $A + B = B + A$



A	B	A + B	=	B	A	B + A
0	0	0		0	0	0
0	1	1		0	1	1
1	0	1		1	0	1
1	1	1		1	1	1

This law can be extended to any number of variables. For example,

$$A + B + C = B + C + A = C + A + B = B + A + C$$

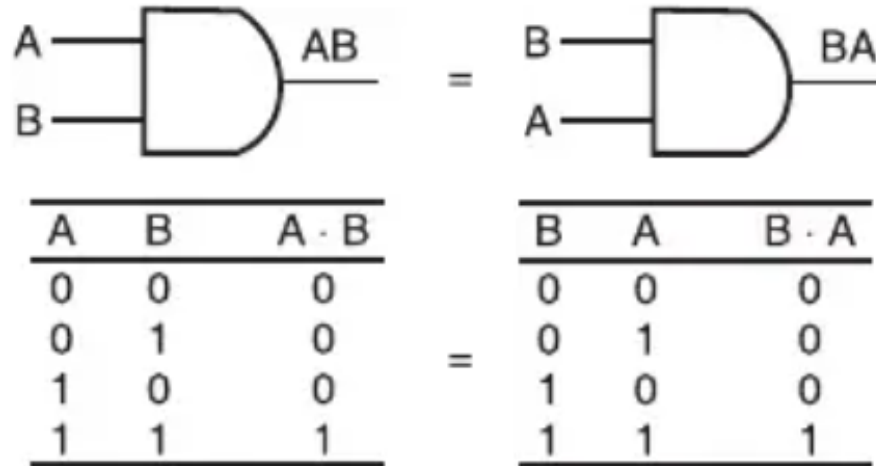
Law 2: $A \cdot B = B \cdot A$

Boolean Algebra



Commutative Laws

Law 2: $A \cdot B = B \cdot A$



This law can be extended to any number of variables. For example,

$$A \cdot B \cdot C = B \cdot C \cdot A = C \cdot A \cdot B = B \cdot A \cdot C$$

Boolean Algebra

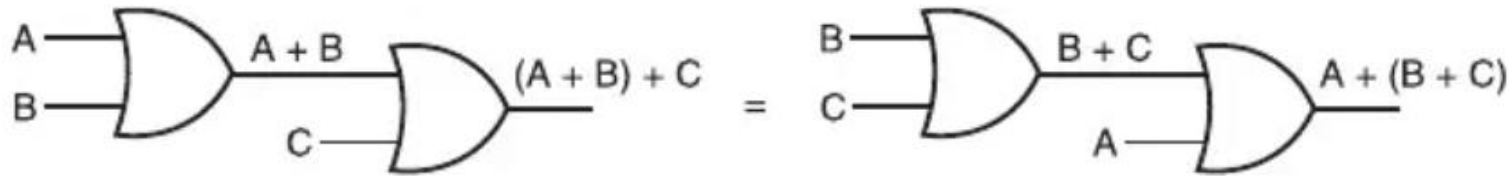


Associative Laws

The associative laws allow grouping of variables. There are two associative laws.

Law 1: $(A + B) + C = A + (B + C)$

A OR B ORed with C is the same as A ORed with B OR C. This law states that the way the variables are grouped and ORed is immaterial. The truth tables given next illustrate this law.



A	B	C	A + B	(A + B) + C
0	0	0	0	0
0	0	1	0	1
0	1	0	1	1
0	1	1	1	1
1	0	0	1	1
1	0	1	1	1
1	1	0	1	1
1	1	1	1	1

=

A	B	C	B + C	A + (B + C)
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	1	1
1	0	0	0	1
1	0	1	1	1
1	1	0	1	1
1	1	1	1	1

This law can be extended to any number of variables. For example,

$$A + (B + C + D) = (A + B + C) + D = (A + B) + (C + D).$$

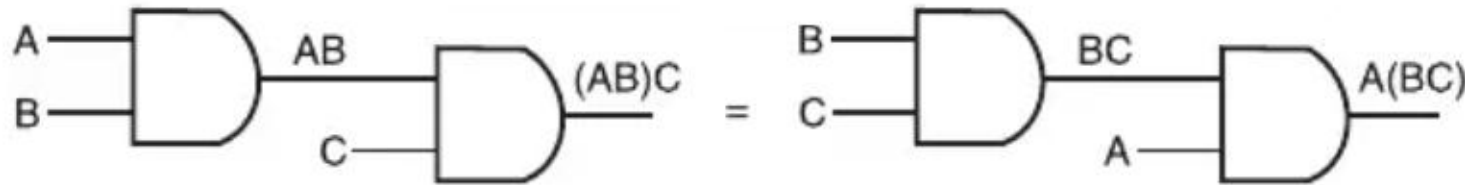
Boolean Algebra



Associative Laws

Law 2: $(A \cdot B)C = A(B \cdot C)$

A AND B ANDed with C is the same as A ANDed with B AND C. This law states that the way the variables are grouped and ANDed is immaterial. See the truth tables below:



A	B	C	AB	(AB)C
0	0	0	0	0
0	0	1	0	0
0	1	0	0	0
0	1	1	0	0
1	0	0	0	0
1	0	1	0	0
1	1	0	1	0
1	1	1	1	1

=

A	B	C	BC	A(BC)
0	0	0	0	0
0	0	1	0	0
0	1	0	0	0
0	1	1	1	0
1	0	0	0	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

This law can be extended to any number of variables. For example,

$$A(BCD) = (ABC)D = (AB)(CD)$$

Boolean Algebra

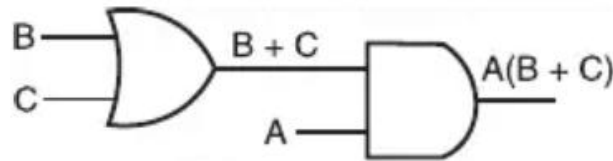


Distributive Laws

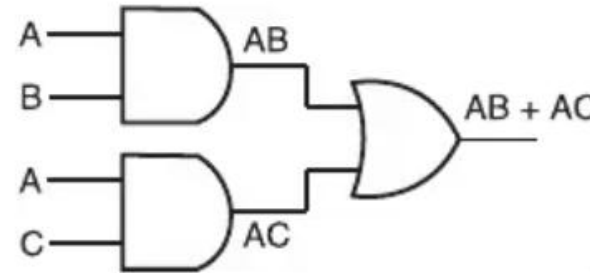
The distributive laws allow factoring or multiplying out of expressions. There are two distributive laws.

Law 1:

$$A(B + C) = AB + AC$$



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A	B	C	B + C	A(B + C)
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	1	0
1	0	0	0	0
1	0	1	1	1
1	1	0	1	1
1	1	1	1	1

=

A	B	C	AB	AC	AB + AC
0	0	0	0	0	0
0	0	1	0	0	0
0	1	0	0	0	0
0	1	1	0	0	0
1	0	0	0	0	0
1	0	1	0	1	1
1	1	0	1	0	1
1	1	1	1	1	1

This law applies to single variables as well as combinations of variables. For example,

$$ABC(D + E) = ABCD + ABCE$$

$$AB(CD + EF) = ABCD + ABEF$$

Boolean Algebra



Distributive Laws

Law 2: $A + BC = (A + B)(A + C)$

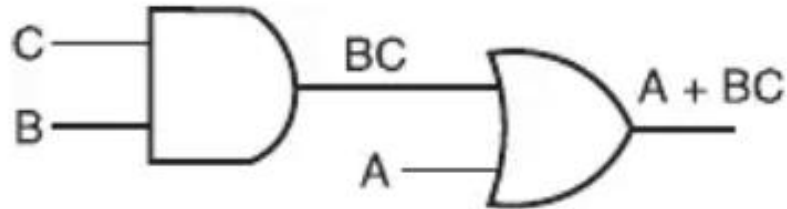
This law states that ANDing of several variables and ORing the result with a single variable is equivalent to ORing that single variable with each of the several variables and then ANDing the sums. This can be proved algebraically as shown below. Also, the truth tables given next illustrate this law.

$$\begin{aligned}\text{RHS} &= (A + B)(A + C) \\ &= AA + AC + BA + BC \\ &= A + AC + AB + BC \\ &= A(1 + C + B) + BC \\ &= A \cdot 1 + BC & (\because 1 + C + B = 1 + B = 1) \\ &= A + BC \\ &= \text{LHS}\end{aligned}$$

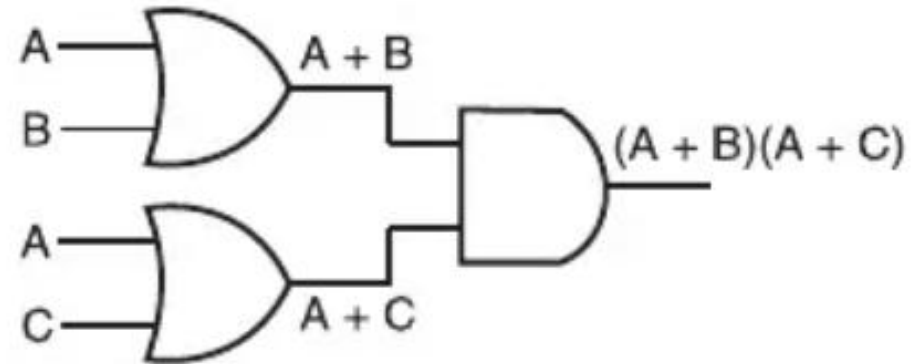
Boolean Algebra



Distributive Laws



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A	B	C	BC	A + BC
0	0	0	0	0
0	0	1	0	0
0	1	0	0	0
0	1	1	1	1
1	0	0	0	1
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

=

A	B	C	A + B	A + C	(A + B)(A + C)
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	1	0	0
0	1	1	1	1	1
1	0	0	1	1	1
1	0	1	1	1	1
1	1	0	1	1	1
1	1	1	1	1	1

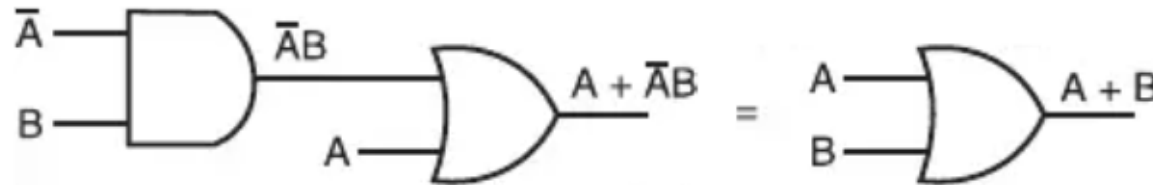
Boolean Algebra



Redundant Literal Rule

Law 1: $A + \bar{A}B = A + B$

This law states that ORing of a variable with the AND of the complement of that variable with another variable, is equal to the ORing of the two variables. See the truth tables given below.



A	B	$\bar{A}B$	$A + \bar{A}B$
0	0	0	0
0	1	1	1
1	0	0	1
1	1	0	1

A	B	$A + B$
0	0	0
0	1	1
1	0	1
1	1	1

This law can be proved algebraically as shown below.

$$\begin{aligned} A + \bar{A}B &= (A + \bar{A})(A + B) \\ &= 1 \cdot (A + B) \\ &= A + B \end{aligned}$$

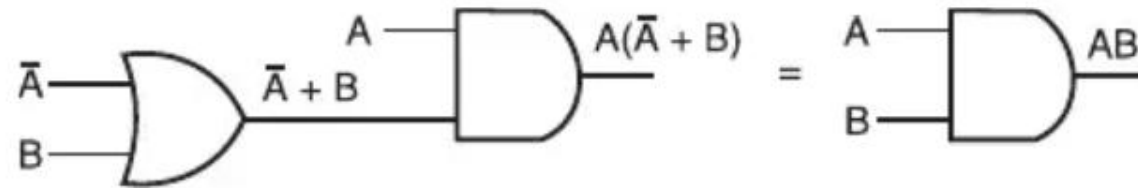
Boolean Algebra



Redundant Literal Rule

Law 2: $A(\bar{A} + B) = AB$

This law states that ANDing of a variable with the OR of the complement of that variable with another variable, is equal to the ANDing of the two variables. See the truth tables given below.



A	B	$\bar{A} + B$	$A(\bar{A} + B)$	=	A	B	AB
0	0	1	0		0	0	0
0	1	1	0		0	1	0
1	0	0	0		1	0	0
1	1	1	1		1	1	1

This law can be proved algebraically as shown below.

$$\begin{aligned} A(\bar{A} + B) &= A\bar{A} + AB \\ &= 0 + AB \\ &= AB \end{aligned}$$

Boolean Algebra



Idempotence Laws

Law 1: $A \cdot A = A$

Idempotence means the same value. We are already familiar with the following laws:

If $A = 0$, then $A \cdot A = 0 \cdot 0 = 0 = A$

If $A = 1$, then $A \cdot A = 1 \cdot 1 = 1 = A$

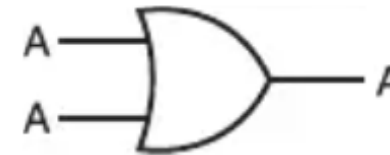


This law states that ANDing of a variable with itself is equal to that variable only.

Law 2: $A + A = A$

If $A = 0$, then $A + A = 0 + 0 = 0 = A$

If $A = 1$, then $A + A = 1 + 1 = 1 = A$



This law states that ORing of a variable with itself is equal to that variable only.

Boolean Algebra

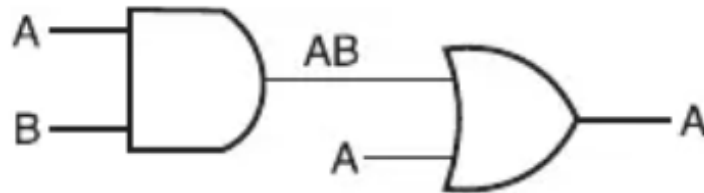


Absorption Laws

There are two laws:

$$\text{Law 1:} \quad A + A \cdot B = A$$

This law states that ORing of a variable (A) with the AND of that variable (A) and another variable (B) is equal to that variable itself (A).



A	B	AB	A + AB
0	0	0	0
0	1	0	0
1	0	0	1
1	1	1	1

Algebraically, we have

$$A + A \cdot B = A(1 + B) = A \cdot 1 = A$$

Therefore,

$$A + A \cdot \text{Any term} = A$$

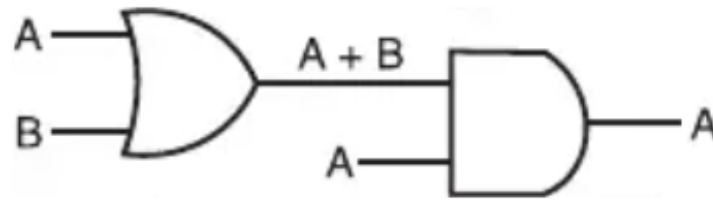
Boolean Algebra



Absorption Laws

Law 2: $A(A + B) = A$

This law states that ANDing of a variable (A) with the OR of that variable (A) and another variable (B) is equal to that variable itself (A).



A	B	A + B	A(A + B)
0	0	0	0
0	1	1	0
1	0	1	1
1	1	1	1

Algebraically, we have

$$A(A + B) = A \cdot A + A \cdot B = A + AB = A(1 + B) = A \cdot 1 = A$$

Therefore,

$$A(A + \text{Any term}) = A$$

Boolean Algebra



Consensus Theorem

Theorem 1: $AB + \bar{A}C + BC = AB + \bar{A}C$

Proof:

$$\begin{aligned}\text{LHS} &= AB + \bar{A}C + BC \\ &= AB + \bar{A}C + BC(A + \bar{A}) \\ &= AB + \bar{A}C + BCA + BC\bar{A} \\ &= AB(1 + C) + \bar{A}C(1 + B) \\ &= AB(1) + \bar{A}C(1) \\ &= AB + \bar{A}C \\ &= \text{RHS}\end{aligned}$$

This theorem can be extended to any number of variables. For example,

$$AB + \bar{A}C + BCD = AB + \bar{A}C$$

$$\text{LHS} = AB + \bar{A}C + BCD = AB + \bar{A}C + BC + BCD = AB + \bar{A}C + BC = AB + \bar{A}C = \text{RHS}$$

Boolean Algebra



Consensus Theorem

Theorem 2: $(A + B)(\bar{A} + C)(B + C) = (A + B)(\bar{A} + C)$

Proof:

$$\begin{aligned}\text{LHS} &= (A + B)(\bar{A} + C)(B + C) = (A\bar{A} + AC + B\bar{A} + BC)(B + C) \\ &= (AC + BC + \bar{A}B)(B + C) \\ &= ABC + BC + \bar{A}B + AC + BC + \bar{A}BC = AC + BC + \bar{A}B \\ \text{RHS} &= (A + B)(\bar{A} + C) \\ &= A\bar{A} + AC + BC + \bar{A}B \\ &= AC + BC + \bar{A}B = \text{LHS}\end{aligned}$$

Boolean Algebra

Transposition Theorem



Theorem: $AB + \bar{A}C = (A + C)(\bar{A} + B)$

Proof:

$$\begin{aligned}\text{RHS} &= (A + C)(\bar{A} + B) \\ &= A\bar{A} + C\bar{A} + AB + CB \\ &= 0 + \bar{A}C + AB + BC \\ &= \bar{A}C + AB + BC(A + \bar{A}) \\ &= AB + ABC + \bar{A}C + \bar{A}BC \\ &= AB + \bar{A}C \\ &= \text{LHS}\end{aligned}$$

Boolean Algebra



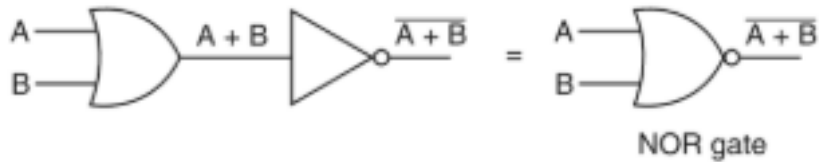
De Morgan's Theorem

De Morgan's Theorem represents two of the most powerful laws in Boolean Algebra:

Law 1: $\overline{A + B} = \bar{A} \cdot \bar{B}$

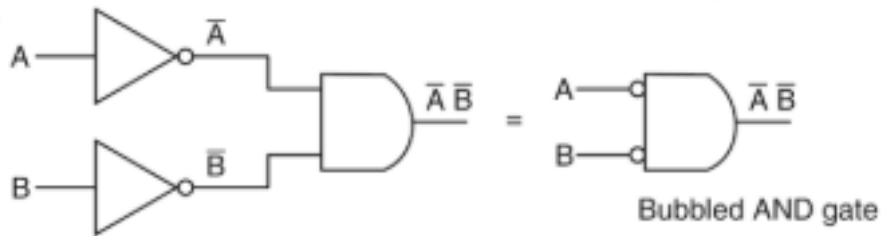
This law states that the complement of a sum of variables is equal to the product of their individual complements.

LHS



NOR gate

RHS



Bubbled AND gate

A	B	A + B	$\overline{A + B}$
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

=

A	B	$\bar{A}$	$\bar{B}$	$\bar{A} \cdot \bar{B}$
0	0	1	1	1
0	1	1	0	0
1	0	0	1	0
1	1	0	0	0

$$\overline{A + B + C + D + \dots} = \bar{A} \bar{B} \bar{C} \bar{D} \dots$$

$$\overline{AB + CD + EFG + \dots} = (\bar{A}\bar{B})(\bar{C}\bar{D})(\bar{E}\bar{F}\bar{G}) \dots = (\bar{A} + \bar{B})(\bar{C} + \bar{D})(\bar{E} + \bar{F} + \bar{G}) \dots$$

Boolean Algebra

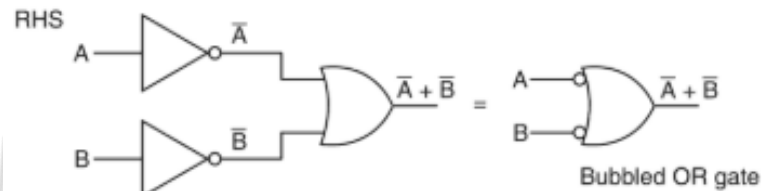
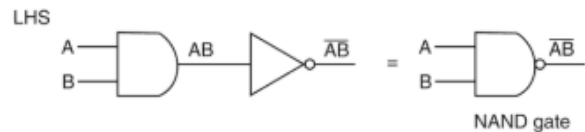


De Morgan's Theorem

De Morgan's Theorem represents two of the most powerful laws in Boolean Algebra:

Law 2: $\overline{A \cdot B} = \overline{A} + \overline{B}$

This law states that the complement of a sum of variables is equal to the product of their individual complements.



A	B	$\overline{AB}$
0	0	1
0	1	1
1	0	1
1	1	0

=

A	B	$\overline{A}$	$\overline{B}$	$\overline{A} + \overline{B}$
0	0	1	1	1
0	1	1	0	1
1	0	0	1	1
1	1	0	0	0

$$\overline{A + B + C + D + \dots} = \overline{A} \overline{B} \overline{C} \overline{D} \dots$$

$$\overline{AB + CD + EFG + \dots} = (\overline{AB})(\overline{CD})(\overline{EFG}) \dots = (\overline{A} + \overline{B})(\overline{C} + \overline{D})(\overline{E} + \overline{F} + \overline{G}) \dots$$

Boolean Algebra



Shannon's Expansion Theorem

Shannon's expansion theorem states that any switching expression can be decomposed with respect to a variable A into two parts; one containing A and the other containing $\bar{A}$. This concept is useful in decomposing complex machines into an interconnection of smaller components.

$$f(A, B, C, \dots) = A \cdot f(1, B, C, \dots) + \bar{A} \cdot f(0, B, C, \dots)$$

$$f(A, B, C, \dots) = [A + f(0, B, C, \dots)] \cdot [\bar{A} + f(1, B, C, \dots)]$$

Boolean Algebra



Example: Demorganize $f = \overline{(A + \bar{B})(C + \bar{D})}$

Complement the entire function	$= (A + \bar{B})(C + \bar{D})$
Change ANDs to ORs and ORs to ANDs	$= A \cdot \bar{B} + C \cdot \bar{D}$
Complement the variables	$= \bar{A} \cdot B + \bar{C} \cdot D$

Example: Apply Demorgan's theorem to the expression $f = \overline{\overline{AB}(CD + \bar{E}F)(\overline{AB} + \overline{CD})}$

The given expression is

$$\begin{aligned} f &= \overline{\overline{AB}(CD + \bar{E}F)(\overline{AB} + \overline{CD})} \\ &= \overline{\overline{AB}} + \overline{CD + \bar{E}F} + \overline{\overline{AB} + \overline{CD}} \\ &= AB + (\overline{CD} \cdot \overline{\bar{E}F}) + (\overline{\overline{AB}} \cdot \overline{\overline{CD}}) \\ &= AB + (\bar{C} + \bar{D})(E + F) + ABCD \end{aligned}$$

Example: Reduce the expression: $f = \overline{\overline{AB}} + \overline{A} + AB$

The given expression is

$$f = \overline{\overline{AB}} + \overline{A} + AB$$

Break the upper line between $\overline{\overline{AB}}$ and $\overline{A}$, and that between $\overline{A}$ and AB , and change the signs between $\overline{\overline{AB}}$, $\overline{A}$, AB and simplify.

$$\begin{aligned} f &= \overline{\overline{AB}} \cdot \overline{\overline{A}} \cdot \overline{AB} \\ &= AB \cdot A \cdot \overline{AB} \\ &= AB \cdot \overline{AB} \\ &= 0 \end{aligned}$$

Boolean Algebra



Duality

Given Expression	Dual		
1. $\bar{0} = 1$	$\bar{1} = 0$	15. $(A + B)(\bar{A} + C)(B + C) = (A + B)(\bar{A} + C)$	$AB + \bar{A}C + BC = AB + \bar{A}C$
2. $0 \cdot 1 = 0$	$1 + 0 = 1$	16. $(A + C)(\bar{A} + B) = AB + \bar{A}C$	$AC + \bar{A}B = (A + B)(\bar{A} + C)$
3. $0 \cdot 0 = 0$	$1 + 1 = 1$	17. $A + \bar{B}C = (A + \bar{B})(A + C)$	$A(\bar{B} + C) = (A\bar{B} + AC)$
4. $1 \cdot 1 = 1$	$0 + 0 = 0$	18. $(A + B)(C + D) = AC + AD + BC + BD$	$(AB + CD) = (A + C)(A + D)(B + C)(B + D)$
5. $A \cdot 0 = 0$	$A + 1 = 1$	19. $A + B = AB + \bar{A}B + A\bar{B}$	$AB = (A + B)(\bar{A} + B)(A + \bar{B})$
6. $A \cdot 1 = A$	$A + 0 = A$	20. $A + B(\overline{C + DE}) = A + B\bar{C}DE$	$A[B + (\overline{C \cdot D + E})] = A \cdot (B + \bar{C} + D + E)$
7. $A \cdot A = A$	$A + A = A$	21. $\overline{\bar{A}B + \bar{A} + AB} = 0$	$\overline{A + B \cdot \bar{A} \cdot (A + B)} = 1$
8. $A \cdot \bar{A} = 0$	$A + \bar{A} = 1$	22. $AB + \bar{A}C + A\bar{B}C(AB + C) = 1$	$(A + B)(\bar{A} + C) \cdot [(A + \bar{B} + C) + (A + B)C] = 0$
9. $A \cdot B = B \cdot A$	$A + B = B + A$	23. $ABD + ABCD = ABD$	$(A + B + D)(A + B + C + D) = (A + B + D)$
10. $A \cdot (B \cdot C) = (A \cdot B) \cdot C$	$A + (B + C) = (A + B) + C$	24. $\overline{\bar{A}B + ABC} + A(B + A\bar{B}) = 0$	$\overline{(A + \bar{B}) \cdot (A + B + C)} \cdot (A + [B(A + \bar{B})]) = 1$
11. $A \cdot (B + C) = AB + AC$	$A + BC = (A + B)(A + C)$	25. $A + \bar{B}C(A + \bar{B}C) = A + \bar{B}C$	$A \cdot [(\bar{B} + C) + A \cdot (\bar{B} + C)] = A \cdot (\bar{B} + C)$
12. $A(A + B) = A$	$A + AB = A$		
13. $A \cdot (A \cdot B) = A \cdot B$	$A + A + B = A + B$		
14. $\overline{AB} = \bar{A} + \bar{B}$	$\overline{A + B} = \bar{A}\bar{B}$		

Boolean Algebra



Reduction of Boolean Expressions

- (a) Multiply all variables necessary to remove parentheses.
(b) Look for identical terms. Only one of those terms be retained and all others dropped. For example,

$$AB + AB + AB + AB = AB$$

- (c) Look for a variable and its negation in the same term. This term can be dropped. For example,

$$A \cdot B\bar{B} = A \cdot 0 = 0; \quad ABC\bar{C} = AB \cdot 0 = 0$$

- (d) Look for pairs of terms that are identical except for one variable which may be missing in one of the terms. The larger term can be dropped. For example,

$$AB\bar{C}\bar{D} + ABC\bar{C} = AB\bar{C}(\bar{D} + 1) = AB\bar{C} \cdot 1 = AB\bar{C}$$

- (e) Look for pairs of terms which have the same variables, with one or more variables complemented. If a variable in one term of such a pair is complemented while in the second term it is not, then such terms can be combined into a single term with that variable dropped. For example,

$$AB\bar{C}\bar{D} + ABC\bar{D} = AB\bar{C}(\bar{D} + D) = AB\bar{C} \cdot 1 = AB\bar{C}$$

$$AB(C + D) + AB(\bar{C} + \bar{D}) = AB[(C + D) + (\bar{C} + \bar{D})] = AB \cdot 1 = AB$$

Example: Reduce the expression: $f = A[B + \bar{C}(AB + AC)]$

The given expression is

Demorganize $\overline{AB + AC}$

Demorganize $\overline{AB}$ and $\overline{AC}$

Multiply $(\bar{A} + \bar{B})(\bar{A} + C)$

Simplify

Simplify

Simplify

Simplify

Simplify

$$f = A[B + \bar{C}(\overline{AB + AC})]$$

$$= A[B + \bar{C}(\bar{A}\bar{B} \cdot \bar{A}\bar{C})]$$

$$= A[B + \bar{C}(\bar{A} + \bar{B})(\bar{A} + C)]$$

$$= A[B + \bar{C}(\bar{A}\bar{A} + \bar{A}\bar{C} + \bar{B}\bar{A} + \bar{B}C)]$$

$$= A(B + \bar{C}\bar{A} + \bar{C}\bar{A}\bar{C} + \bar{C}\bar{B}\bar{A} + \bar{C}\bar{B}C)$$

$$= A(B + \bar{C}\bar{A} + 0 + \bar{C}\bar{B}\bar{A} + 0)$$

$$= AB + A\bar{C}\bar{A} + A\bar{C}\bar{B}\bar{A}$$

$$= AB + 0 + 0$$

$$= AB$$

Boolean Algebra



Example: Reduce the expression: $f = A + B[AC + (B + \bar{C})D]$

The given expression is

Expand $(B + \bar{C})D$

Expand $B(AC + BD + \bar{C}D)$

Write in order

Factor

Reduce

Simplify

$$\begin{aligned}f &= A + B[AC + (B + \bar{C})D] \\&= A + B(AC + BD + \bar{C}D) \\&= A + BAC + BBD + B\bar{C}D \\&= A + ABC + BD + B\bar{C}D \\&= A(1 + BC) + BD(1 + \bar{C}) \\&= A \cdot 1 + BD \cdot 1 \\&= A + BD\end{aligned}$$

Example: Reduce the expression: $f = \overline{(A + \overline{BC})}(A\bar{B} + ABC)$

Demorganize $\overline{(A + \overline{BC})}$

Simplify

Multiply

Rearrange

$$\begin{aligned}&= (\overline{\overline{A}}\overline{\overline{BC}})(A\bar{B} + ABC) \\&= (\overline{A}BC)(A\bar{B} + ABC) \\&= \overline{A}BC A\bar{B} + \overline{A}BC ABC \\&= A\bar{A}B\bar{B}C + A\bar{A}BBCC \\&= 0 + 0 = 0\end{aligned}$$

Boolean Algebra



Example: Reduce the expression: $f = (B + BC)(B + \bar{B}C)(B + D)$

The given expression is

Multiply the first two terms

Reduce

Factor

Reduce

Expand

Simplify

$$\begin{aligned}f &= (B + BC)(B + \bar{B}C)(B + D) \\&= (BB + BCB + B\bar{B}C + BC\bar{B}C)(B + D) \\&= (B + BC + 0 + 0)(B + D) \\&= B(1 + C)(B + D) \\&= B(B + D) \\&= BB + BD \\&= B(1 + D) = B\end{aligned}$$

Example: Reduce the expression: Show that $AB + A\bar{B}C + B\bar{C}$

Solution

$$\begin{aligned}AB + A\bar{B}C + B\bar{C} &= A(B + \bar{B}C) + B\bar{C} \\&= A(B + \bar{B})(B + C) + B\bar{C} \\&= AB + AC + B\bar{C} \\&= AB(C + \bar{C}) + AC + B\bar{C} \\&= ABC + AB\bar{C} + AC + B\bar{C} \\&= AC(1 + B) + B\bar{C}(1 + A) \\&= AC + B\bar{C}\end{aligned}$$

Boolean Algebra



Example: Reduce the expression: $f = A\bar{B}C + B + B\bar{D} + AB\bar{D} + \bar{A}C$

Solution

$$\begin{aligned} A\bar{B}C + B + B\bar{D} + AB\bar{D} + \bar{A}C &= A\bar{B}C + \bar{A}C + B(1 + \bar{D} + A\bar{D}) \\ &= C(\bar{A} + A\bar{B}) + B \\ &= C(\bar{A} + A)(\bar{A} + \bar{B}) + B \\ &= C\bar{A} + C\bar{B} + B \\ &= (B + C)(B + \bar{B}) + C\bar{A} \\ &= B + C + C\bar{A} \\ &= B + C(1 + \bar{A}) \\ &= B + C \end{aligned}$$

Example: Simplify the function: $f(A, B, C) = (A + B)(A + \bar{C}) + \bar{A}\bar{B} + \bar{A}C$

Solution

$$\bar{A}\bar{B} = \overline{A + B}$$

Applying redundant literal rule (RLR) to the 1st and 2nd terms of f

$$f = A + B\bar{C} + \bar{A}\bar{B} + \bar{A}C = A + \bar{A}\bar{B} + \bar{A}C + B\bar{C}$$

Applying RLR to 1st and 2nd terms, and 1st and 3rd terms

$$f = A + \bar{B} + A + \bar{C} + B\bar{C}$$

Applying Idempotence law to 1st and 3rd terms

$$f = A + \bar{B} + \bar{C} + B\bar{C}$$

Applying RLR to 2nd and 4th terms

$$f = A + \bar{B} + \bar{C} + \bar{C}$$

Applying idempotence law to the 3rd and 4th terms

$$f = A + \bar{B} + \bar{C}$$

Boolean Algebra



Computation of Total GATE Inputs

EXAMPLE: How many gate inputs are required to realize the following expressions?

(a) $f_1 = ABC + ABCD + E\bar{F} + AD$

(a)	Write the expression	$ABC + A\bar{B}CD + E\bar{F} + AD$
	Count the AND inputs	$3 + 4 + 2 + 2 = 11$
	Count the AND gates feeding the OR gate	$1 + 1 + 1 + 1 = 4$
	Total gate inputs	$= 15$

(b) $f_2 = A(B + C + \bar{D})(\bar{B} + C + \bar{E})(A + \bar{B} + C + E)$

(b)	Write the expression	$A \cdot (B + C + \bar{D}) \cdot (\bar{B} + C + \bar{E}) \cdot (A + \bar{B} + C + E)$
	Count the OR inputs	$0 + 3 + 3 + 4 = 10$
	Count the OR gates feeding the AND gate	$1 + 1 + 1 + 1 = 4$
	Total gate inputs	$= 14$

Boolean Algebra



Boolean Expressions and Logic Diagrams

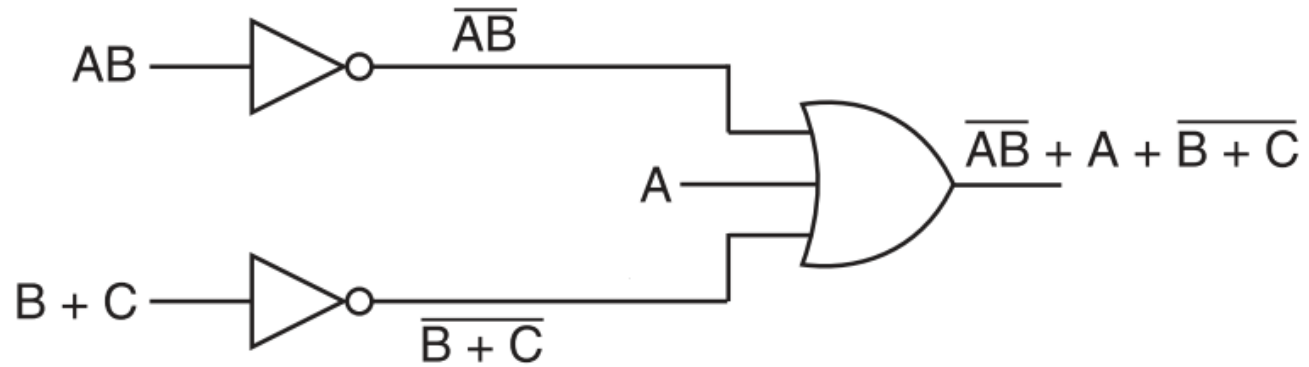
Converting Boolean Expressions to Logic

start with the output and work towards the input

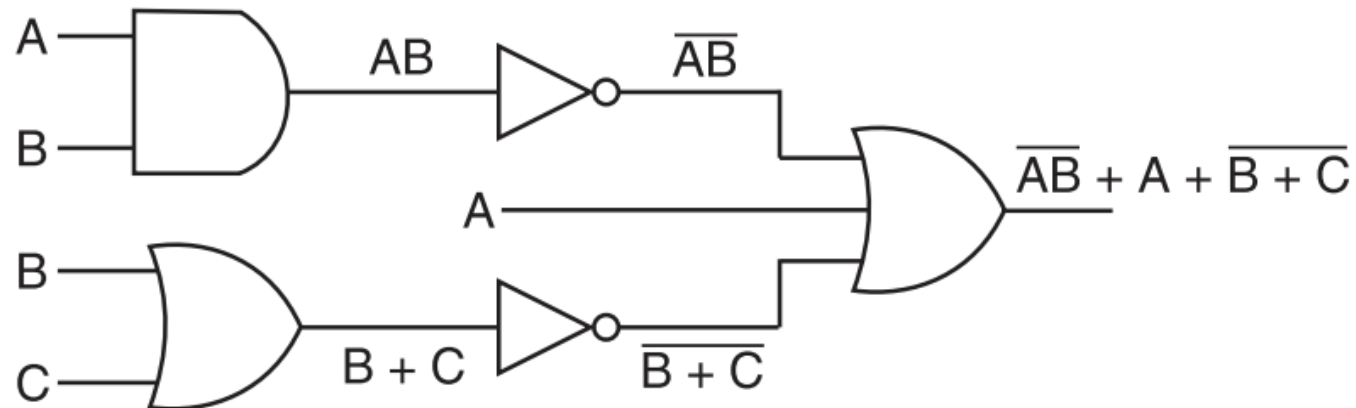
$$\overline{AB} + A + \overline{B + C}$$



Step-1



Step-2



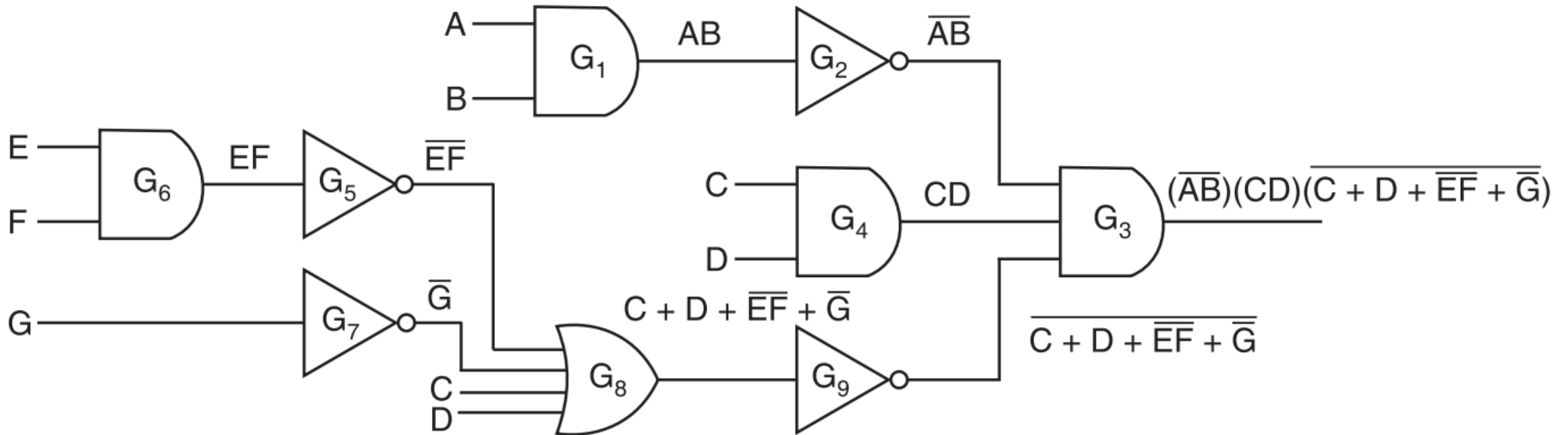
Step-3

Boolean Algebra



Converting Logic to Boolean Expressions

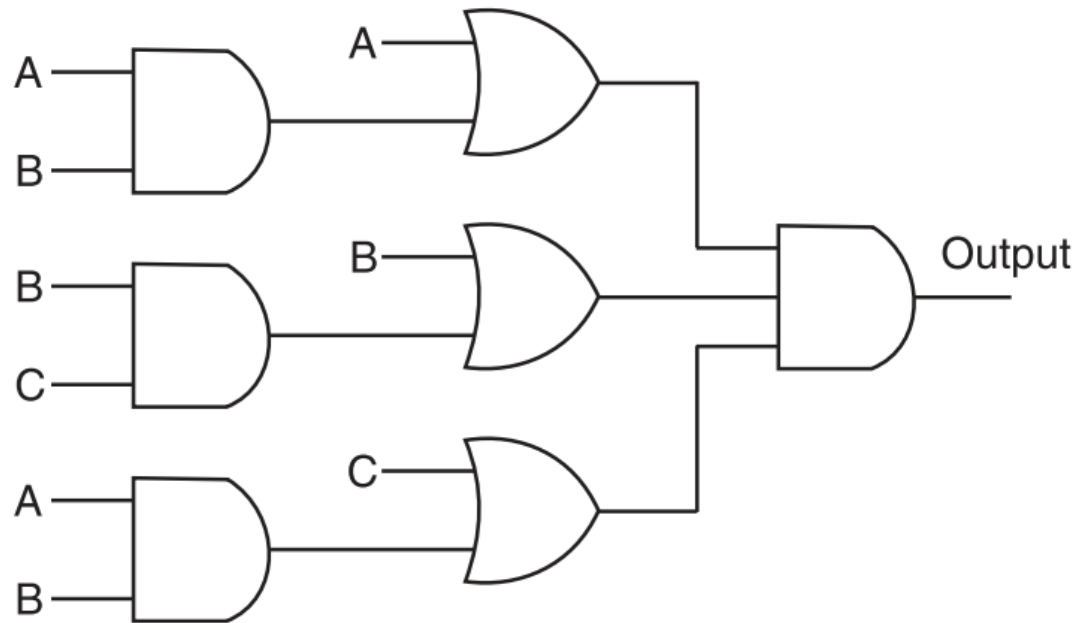
start with the input signals and develop the terms of the Boolean expression until the output is reached.



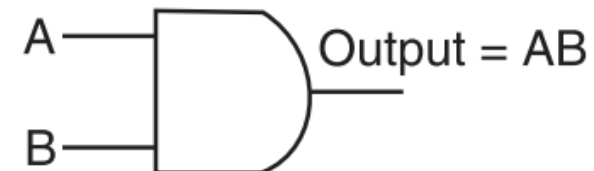
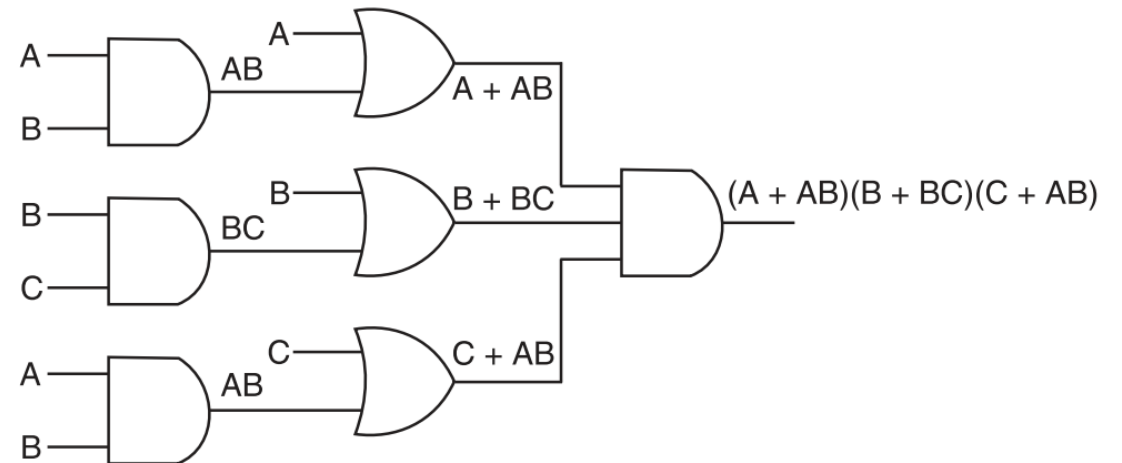
Boolean Algebra



EXAMPLE: Write the Boolean expression for the logic diagram given below and simplify it as much as possible and draw the logic diagram that implements the simplified expression.



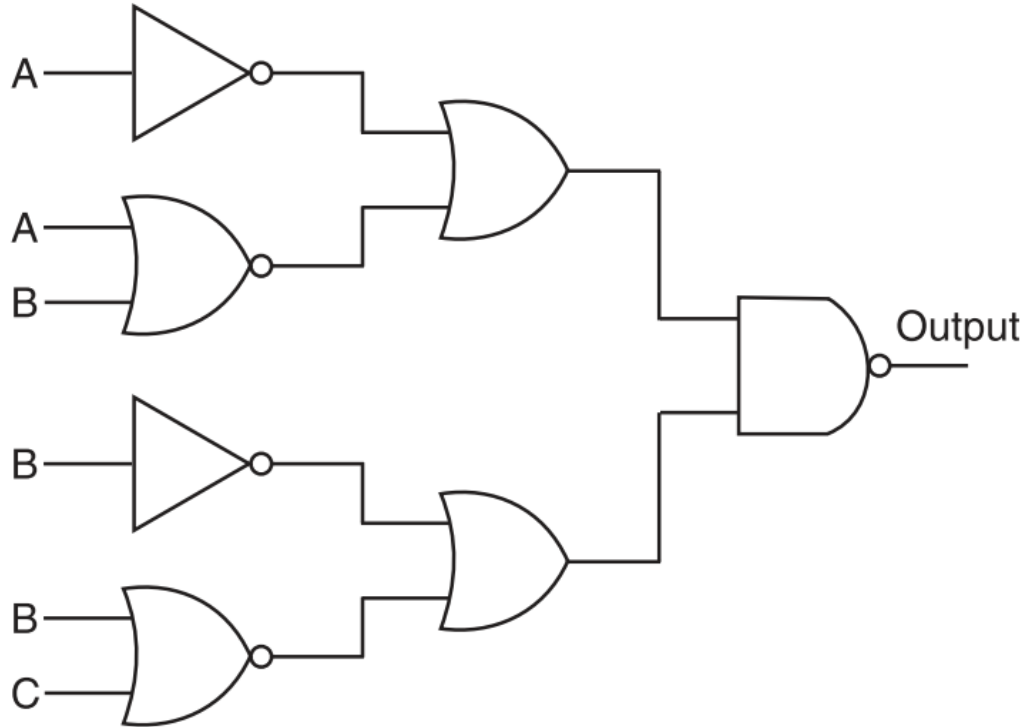
$$\begin{aligned}\text{Output} &= (A + AB)(B + BC)(C + AB) \\ &= A(1 + B)B(1 + C)(C + AB) \\ &= AB(C + AB) \\ &= ABC + AB \\ &= AB(1 + C) \\ &= AB\end{aligned}$$



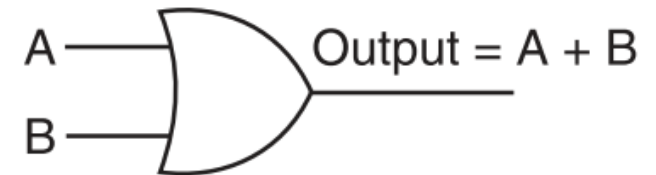
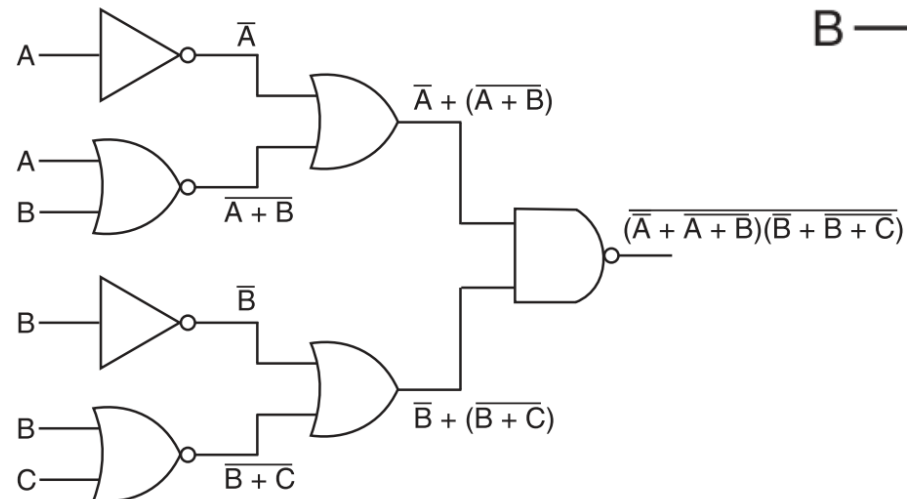
Boolean Algebra



EXAMPLE: Draw the simplest possible logic diagram that implements the output of the logic diagram shown below.

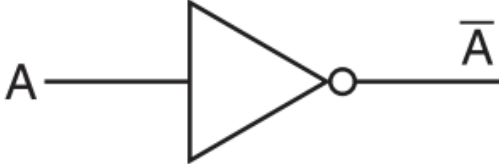
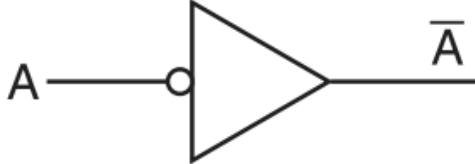
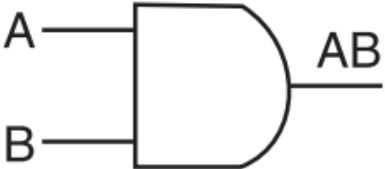
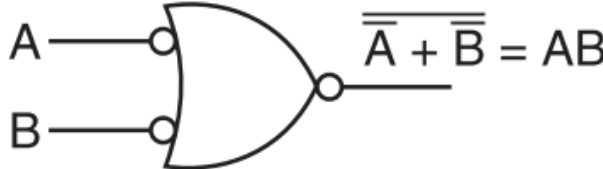
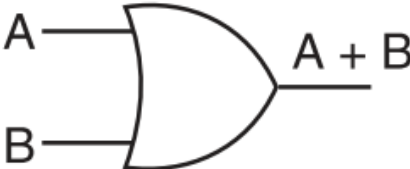
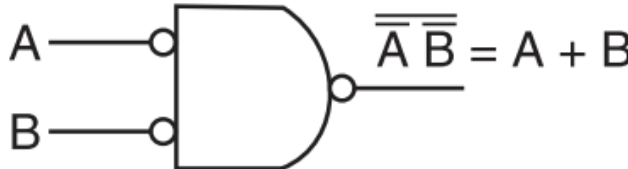
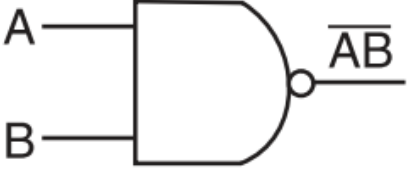
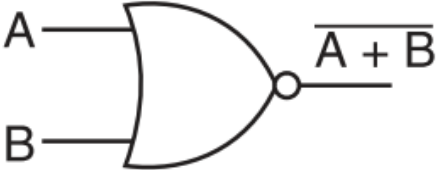


$$\begin{aligned}
 \text{Output} &= \overline{(\overline{A} + \overline{A} + B)}(\overline{B} + \overline{B} + C) \\
 &= \overline{(\overline{A} + \overline{A} + B)} + \overline{(\overline{B} + \overline{B} + C)} \\
 &= \overline{\overline{A}} \cdot \overline{(\overline{A} + B)} + \overline{\overline{B}} \cdot \overline{(\overline{B} + C)} \\
 &= A \cdot (A + B) + B \cdot (B + C) \\
 &= AA + AB + B + BC \\
 &= A + AB + B + BC \\
 &= A(1 + B) + B(1 + C) \\
 &= A + B
 \end{aligned}$$



Boolean Algebra



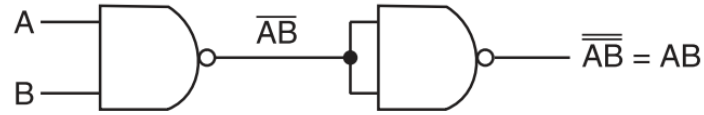
	Traditional symbols Active HIGH	Traditional symbols Active LOW
NOT Gate		
AND Gate		
OR Gate		
NAND Gate		
NOR Gate		

Boolean Algebra



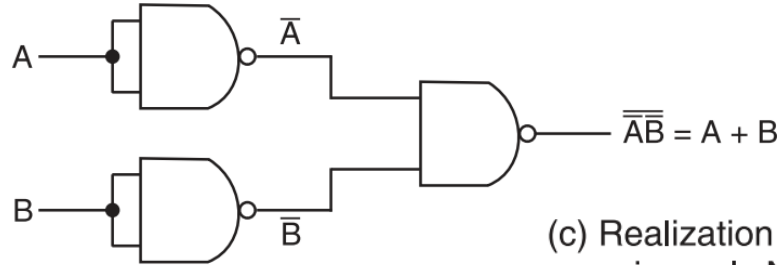
(a) Realization of AND function using only NAND gates

$$AB = \overline{\overline{AB}} = \overline{(\overline{AB})}$$



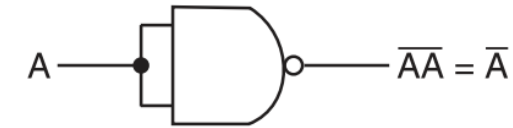
(b) Realization of OR function using only NAND gates

$$A + B = \overline{\overline{A + B}} = \overline{\overline{A} \cdot \overline{B}}$$



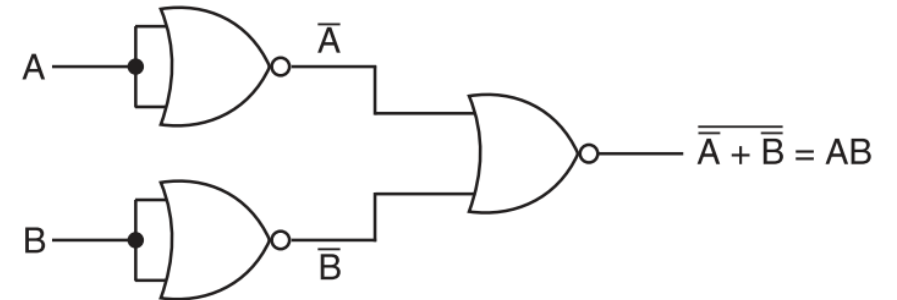
(c) Realization of NOT function using only NAND gates

$$\overline{A} = \overline{A \cdot A}$$



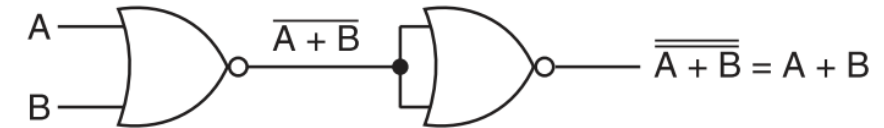
(d) Realization of AND function using only NOR gates

$$AB = \overline{\overline{AB}} = \overline{\overline{A} + \overline{B}}$$



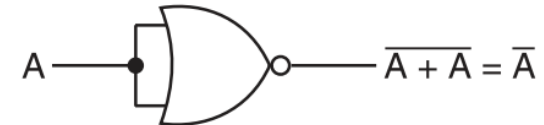
(e) Realization of OR function using only NOR gates

$$A + B = \overline{\overline{A + B}} = \overline{(\overline{A + B})}$$



(f) Realization of NOT function using only NOR gates

$$\overline{A} = \overline{A + A}$$



Boolean Algebra



EXAMPLE: Find the values of the two-valued variables A, B, C, and D by solving the set of simultaneous equations.

$$\bar{A} + AB = 0$$

$$AB = AC$$

$$AB + A\bar{C} + CD = \bar{C}D$$

Given

$$\bar{A} + AB = 0$$

or

$$(\bar{A} + A)(\bar{A} + B) = 0$$

or

$$(1)(\bar{A} + B) = 0$$

or

$$\bar{A} + B = 0$$

i.e. $\bar{A}$ must be 0 and B must be 0. Therefore, A = 1 and B = 0.

Also

$$AB = AC$$

or

$$1 \cdot 0 = 1 \cdot C$$

Therefore

$$C = 0$$

Also,

$$AB + A\bar{C} + CD = \bar{C}D$$

or

$$1 \cdot 0 + 1 \cdot 1 + 0 \cdot D = 1 \cdot D$$

or

$$1 = D$$

Therefore, the values of A, B, C, and D are A = 1, B = 0, C = 0, and D = 1.

Boolean Algebra



EXAMPLE: Simplify the following Boolean expressions to a minimum number of literals

(a) $\bar{x}\bar{y} + xy + \bar{x}y$

(a) $\bar{x}\bar{y} + xy + \bar{x}y = \bar{x}\bar{y} + y(x + \bar{x}) = \bar{x}\bar{y} + y = (y + \bar{y})(y + \bar{x}) = \bar{x} + y$

(b) $x\bar{y} + \bar{y}\bar{z} + \bar{x}\bar{z}$

(b) $x\bar{y} + \bar{y}\bar{z} + \bar{x}\bar{z} = x\bar{y} + \bar{x}\bar{z} + \bar{y}\bar{z}(x + \bar{x}) = x\bar{y} + \bar{x}\bar{z} + \bar{y}\bar{z}x + \bar{y}\bar{z}\bar{x}$
 $= x\bar{y}(1 + \bar{z}) + \bar{x}\bar{z}(1 + \bar{y}) = x\bar{y} + \bar{x}\bar{z}$

(n) $ABC\bar{D} + A + AB\bar{D} + (\bar{D})(\bar{A}\bar{B}\bar{C})$

(n) $ABC\bar{D} + A + AB\bar{D} + (\bar{D})(\bar{A}\bar{B}\bar{C}) = ABC\bar{D} + A + AB\bar{D} + (\bar{A}\bar{B}\bar{C}\bar{D})$
 $= A(1 + B\bar{D} + B\bar{C}\bar{D}) + \bar{A}\bar{B}\bar{C}\bar{D} = A + \bar{A}\bar{B}\bar{C}\bar{D}$
 $= (A + \bar{A})(A + \bar{B}\bar{C}\bar{D}) = A + \bar{B}\bar{C}\bar{D}$

EXAMPLE:

Given $\overline{AB} + \overline{A}B = C$, find $\overline{AC} + \overline{A}C$.

Solution

$$\begin{aligned}\overline{AC} + \overline{A}C &= \overline{A(\overline{AB} + \overline{A}B)} + \overline{A}(\overline{AB} + \overline{A}B) \\ &= \overline{A(\overline{A} + \overline{B} + \overline{A}B)} + \overline{A}(\overline{A} + \overline{B} + \overline{A}B) \\ &= \overline{A\overline{B}} + \overline{A}[\overline{A}(1 + B) + \overline{B}] \\ &= \overline{A} + B + \overline{A} + \overline{A}\overline{B} = \overline{A} + \overline{A}\overline{B} + B = \overline{A}(1 + \overline{B}) + B = \overline{A} + B\end{aligned}$$

Prove that:

(a) If $A \oplus B = 0$, then $A = B$

(a) The X-OR gate is an anti-coincidence gate. Its output is 0 when both the inputs are equal.
Therefore, if $A \oplus B = 0$, then A must be equal to B.

(b) $A \oplus B = \bar{A} \oplus \bar{B}$

(b)
$$A \oplus B = A\bar{B} + \bar{A}B$$
$$\bar{A} \oplus \bar{B} = \bar{\bar{A}} \cdot \bar{\bar{B}} + \bar{A} \cdot \bar{\bar{B}} = A\bar{B} + \bar{A}B$$
$$\therefore A \oplus B = \bar{A} \oplus \bar{B}$$

(c) $\overline{A \oplus B} = \bar{A} \oplus B = A \oplus \bar{B}$

(c)
$$\bar{A} \oplus B = \bar{\bar{A}} \cdot B + \bar{A} \cdot \bar{\bar{B}} = AB + \bar{A}\bar{B} = \overline{A \oplus B}$$
$$A \oplus \bar{B} = \bar{A} \cdot \bar{\bar{B}} + A \cdot \bar{\bar{B}} = \bar{A}\bar{B} + AB = \overline{A \oplus B}$$
$$\therefore \overline{A \oplus B} = \bar{A} \oplus B = A \oplus \bar{B}$$

Boolean Algebra



Prove that:

(d) $0 \oplus A = A$

(d)
$$0 \oplus A = 0 \cdot \bar{A} + \bar{0} \cdot A = 0 + 1 \cdot A = 0 + A = A$$

(e) $1 \oplus A = \bar{A}$

(e)
$$1 \oplus A = 1 \cdot \bar{A} + \bar{1} \cdot A = \bar{A} + 0 \cdot A = \bar{A} + 0 = \bar{A}$$

Boolean Algebra



Obtain the duals of the following functions:

(a) $\bar{A}B + \bar{A}B\bar{C} + \bar{A}BCD + \bar{A}B\bar{C}\bar{D}E$

(a) $(\bar{A} + B)(\bar{A} + B + \bar{C})(\bar{A} + B + C + D)(\bar{A} + B + \bar{C} + \bar{D} + E)$

(f) $\bar{X}\bar{Y}\bar{Z} + \bar{X}Y\bar{Z} + X\bar{Y}\bar{Z} + X\bar{Y}Z + XY\bar{Z}$

(f) $(\bar{X} + \bar{Y} + \bar{Z})(\bar{X} + Y + \bar{Z})(X + \bar{Y} + \bar{Z})(X + \bar{Y} + Z)(X + Y + \bar{Z})$

(b) $ABEF + AB\bar{E}\bar{F} + \bar{A}\bar{B}EF$

(b) $(A + B + E + F)(A + B + \bar{E} + \bar{F})(\bar{A} + \bar{B} + E + F)$

Boolean Algebra



Find the complements of the following expressions:

$$(a) AB + A(B + C) + \bar{B}(B + D)$$

$$(a) (\bar{A} + \bar{B}) \cdot (\bar{A} + \bar{B} \cdot \bar{C}) \cdot (\bar{B} + \bar{B}\bar{D}) = (\bar{A} + \bar{B}) \cdot (\bar{A} + \bar{B} \cdot \bar{C}) \cdot (B + \bar{B}\bar{D})$$

$$(i) \overline{(ABC)} \overline{(A + B + C)}$$

$$(i) \overline{\overline{(ABC)}} + \overline{\overline{(A + B + C)}} = (ABC) + (A + B + C)$$

$$(f) AB + \overline{AC} + A\bar{B}C$$

$$(f) (\bar{A} + \bar{B}) \cdot (AC) \cdot (\bar{A} + B + \bar{C})$$

Boolean Algebra



Find the complement and the dual of the function given below and then reduce it to a minimum number of literals in each case.

$$f = [(\overline{ab})a][(\overline{ab})b].$$

$$\text{Complement of } [(\overline{ab})a][(\overline{ab})b] = \overline{\overline{ab} + a} + \overline{\overline{ab} + b} = ab + \overline{a} + ab + \overline{b}$$

$$\begin{aligned} \text{(Reduction)} \quad ab + \overline{a} + ab + \overline{b} &= \overline{a} + \overline{b} + ab = (\overline{a} + a)(\overline{a} + b) + \overline{b} \\ &= \overline{a} + b + \overline{b} = \overline{a} + 1 = 1 \end{aligned}$$

$$\text{Dual of } [(\overline{ab})a][(\overline{ab})b] = (\overline{ab} + a) + (\overline{ab} + b)$$

$$\begin{aligned} \text{(Reduction)} \quad (\overline{ab} + a) + (\overline{ab} + b) &= \overline{ab} + a + b = a + b + \overline{ab} = a + b + \overline{a} + \overline{b} \\ &= a + \overline{a} + b + \overline{b} = 1 + 1 = 1 \end{aligned}$$

Boolean Algebra

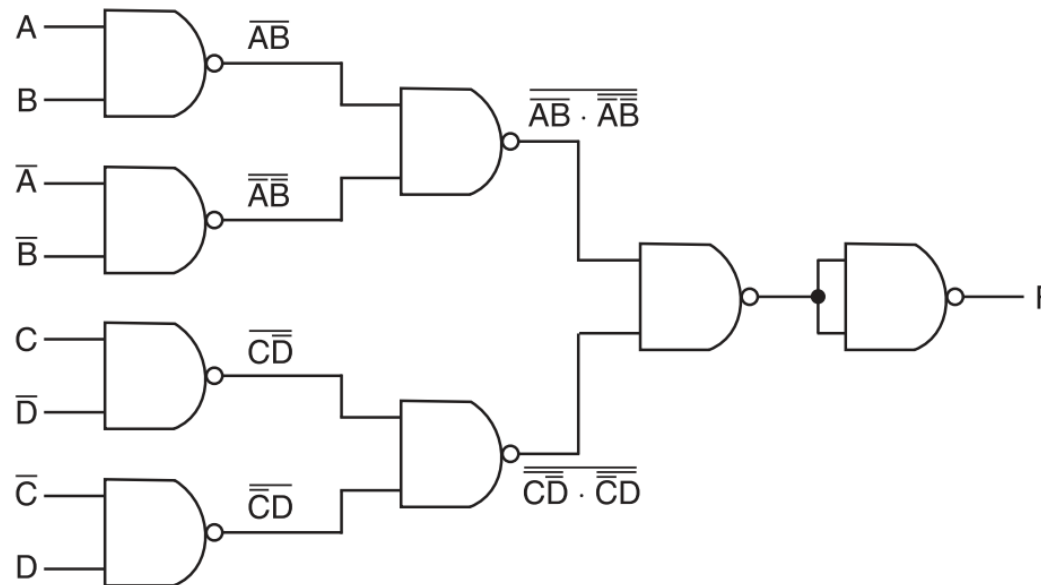


Draw the logic diagram using only two input NAND gates to implement the following expression

$$F = (AB + \bar{A}\bar{B})(C\bar{D} + \bar{C}\bar{D})$$

To implement an expression using only NAND gates, the expression must have only product terms. The given expression in terms of product terms and the corresponding logic diagram are shown below.

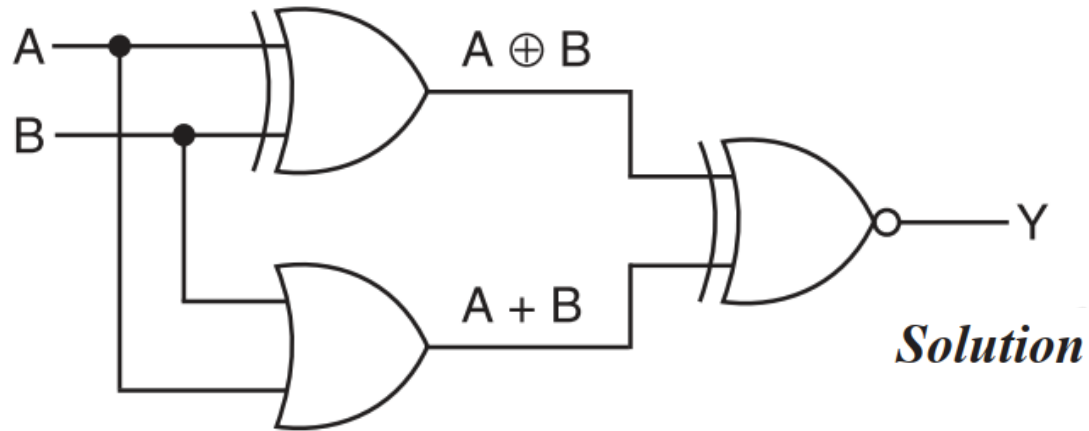
$$F = \overline{\overline{AB + \bar{A}\bar{B}} \cdot \overline{C\bar{D} + \bar{C}\bar{D}}} = \overline{\overline{AB} \cdot \overline{\bar{A}\bar{B}} \cdot \overline{C\bar{D}} \cdot \overline{\bar{C}\bar{D}}}$$



Boolean Algebra

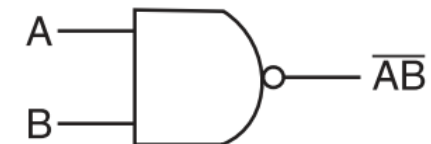


EXAMPLE: Redraw the circuit given in Figure after simplification



$$\begin{aligned} Y &= (A \oplus B) \odot (A + B) \\ &= (A \oplus B)(A + B) + (\overline{A \oplus B})(\overline{A + B}) \\ &= (A\bar{B} + \bar{A}B)(A + B) + (AB + \bar{A}\bar{B})(\bar{A}\bar{B}) \\ &= A\bar{B} + \bar{A}B + \bar{A}\bar{B} = A\bar{B} + \bar{A}(B + \bar{B}) = \bar{A} + A\bar{B} \\ &= (\bar{A} + A)(\bar{A} + \bar{B}) = (\bar{A} + \bar{B}) = \overline{AB} \end{aligned}$$

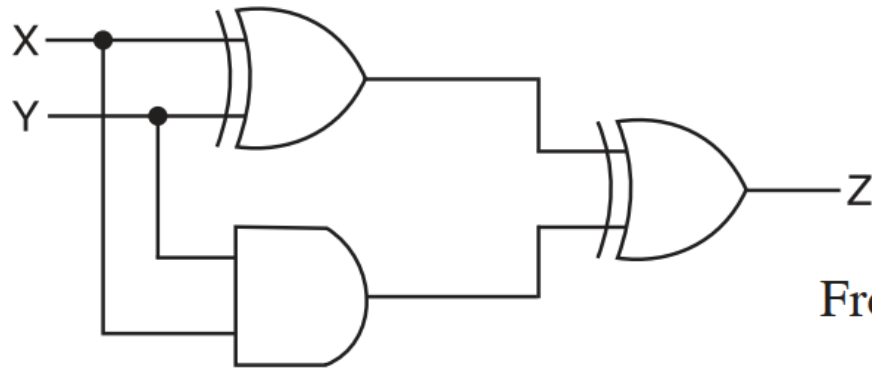
So, the simplified circuit is just a two-input NAND gate.



Boolean Algebra



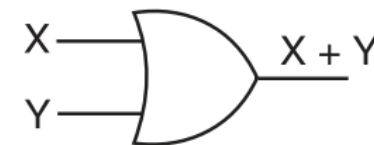
EXAMPLE: Redraw the circuit given after simplification.



From the logic diagram, we see that

$$\begin{aligned} Z &= (X \oplus Y) \oplus (XY) \\ &= (\overline{X \oplus Y}) \cdot XY + (X \oplus Y) \cdot \overline{XY} \\ &= (\overline{X} \overline{Y} + XY)XY + (X \overline{Y} + \overline{X} Y)(\overline{X} + \overline{Y}) \\ &= XY + X \overline{Y} + \overline{X} Y = X(Y + \overline{Y}) + \overline{X} Y = X + \overline{X} Y \\ &= (X + \overline{X})(X + Y) = X + Y \end{aligned}$$

So, the simplified circuit is a two input OR gate.



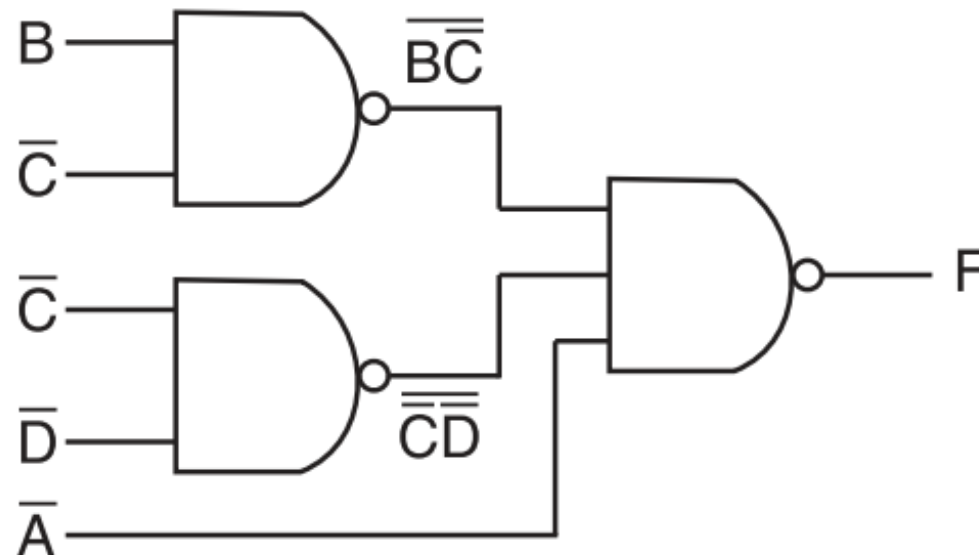
Boolean Algebra



EXAMPLE: Simplify the following expressions and implement them with NAND gate circuits

(a) $F = A\bar{B} + ABD + AB\bar{D} + \bar{A}\bar{C}\bar{D} + \bar{A}B\bar{C}$

(a)
$$\begin{aligned} F &= A\bar{B} + ABD + AB\bar{D} + \bar{A}\bar{C}\bar{D} + \bar{A}B\bar{C} = A\bar{B} + AB(D + \bar{D}) + \bar{A}\bar{C}\bar{D} + \bar{A}B\bar{C} \\ &= A\bar{B} + AB + \bar{A}\bar{C}\bar{D} + \bar{A}B\bar{C} = A(\bar{B} + B) + \bar{A}\bar{C}\bar{D} + \bar{A}B\bar{C} \\ &= A + \bar{A}\bar{C}\bar{D} + \bar{A}B\bar{C} = (A + \bar{A})(A + \bar{C}\bar{D}) + \bar{A}B\bar{C} = A + \bar{C}\bar{D} + \bar{A}B\bar{C} \\ &= (A + \bar{A})(A + B\bar{C}) + \bar{C}\bar{D} = (A + B\bar{C}) + \bar{C}\bar{D} = \overline{\bar{A} \cdot \bar{B}\bar{C} \cdot \bar{C}\bar{D}} \end{aligned}$$



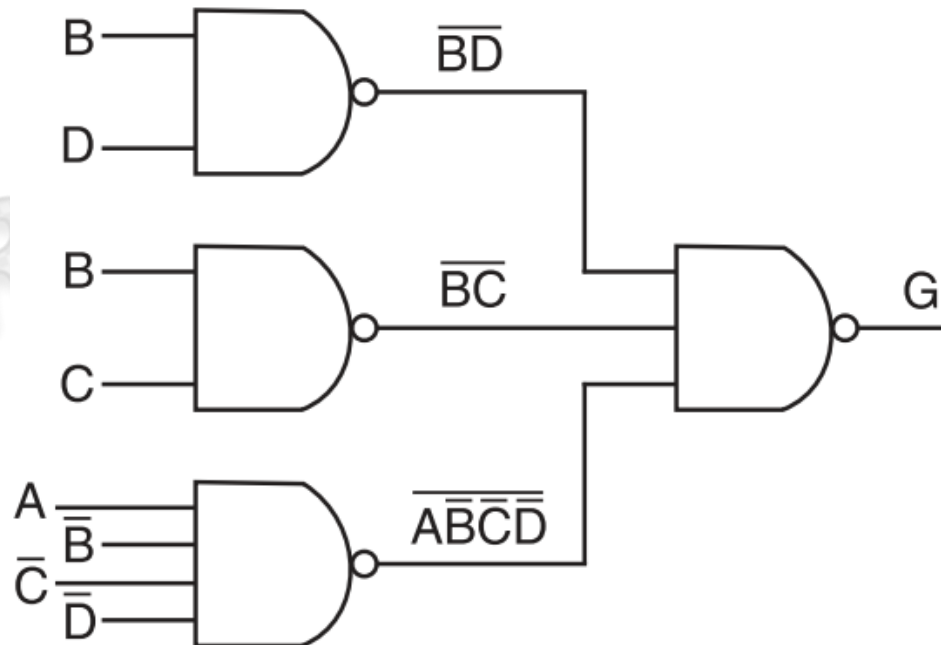
Boolean Algebra



EXAMPLE: Simplify the following expressions and implement them with NAND gate circuits

(b) $G = BD + BCD + A\bar{B}\bar{C}\bar{D}$

$$\begin{aligned} G &= BD + BCD + A\bar{B}\bar{C}\bar{D} = B(D + CD) + A\bar{B}\bar{C}\bar{D} = B(D + C)(D + \bar{D}) + A\bar{B}\bar{C}\bar{D} \\ &= BD + BC + A\bar{B}\bar{C}\bar{D} = \overline{\overline{BD} \cdot \overline{BC} \cdot \overline{A\bar{B}\bar{C}\bar{D}}} \end{aligned}$$

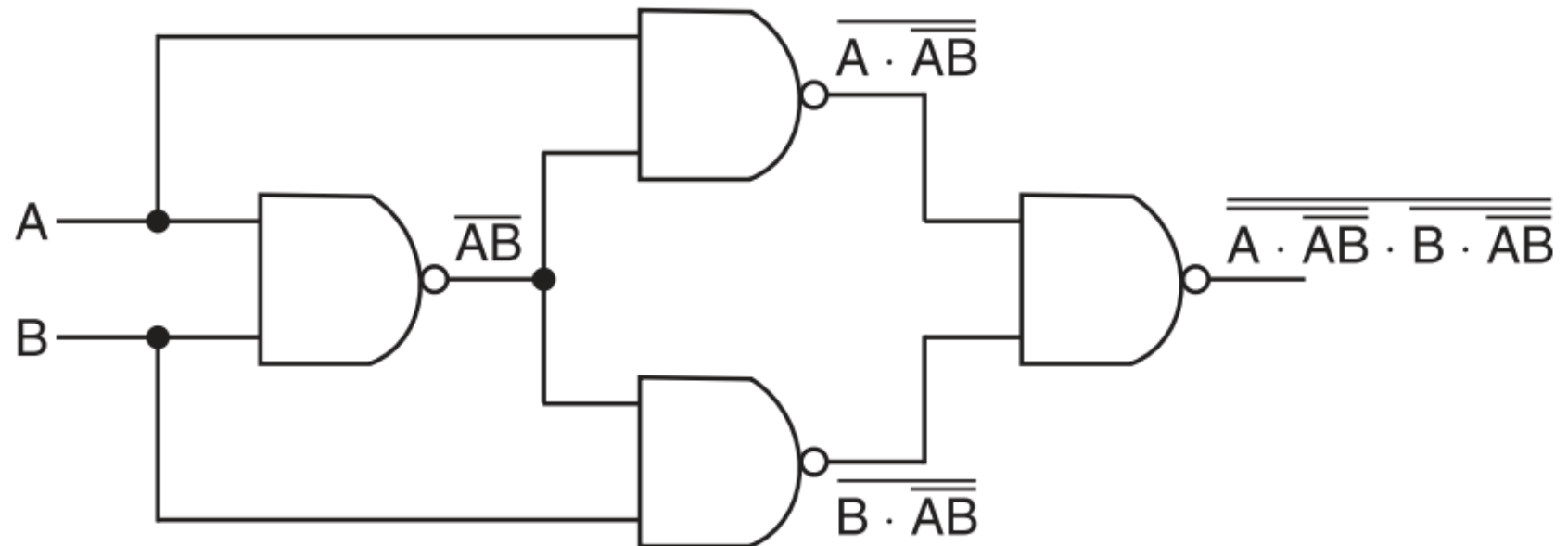


Boolean Algebra



EXAMPLE: Derive the Boolean expression for a two-input Ex-OR gate to realize with two input NAND gates without using complemented variables and draw the circuit.

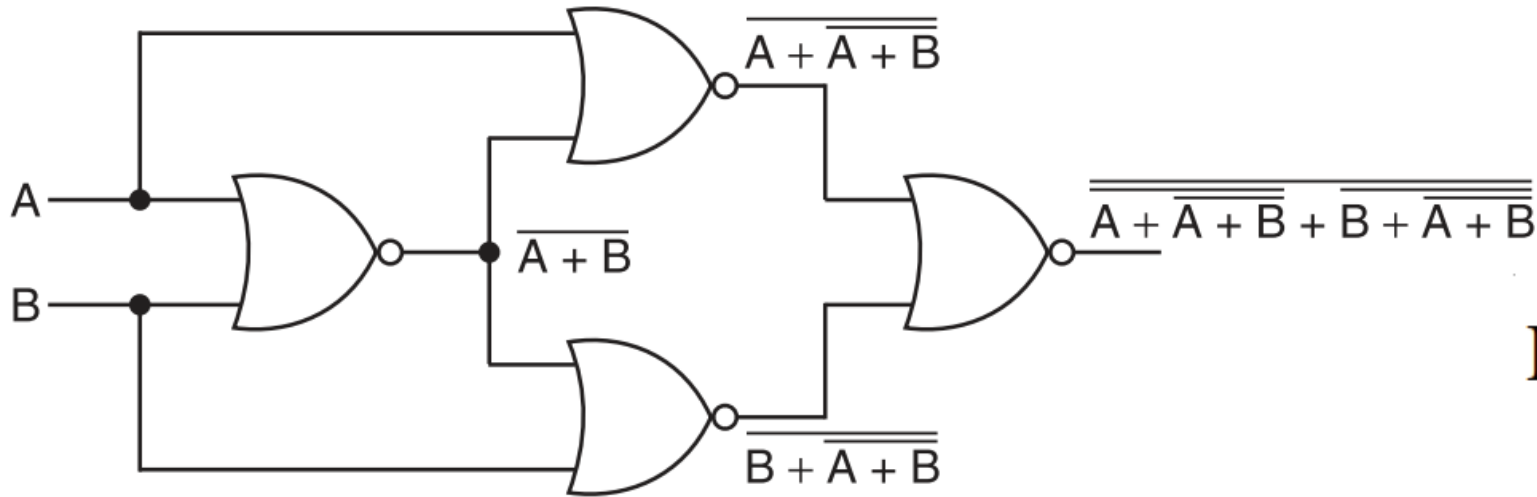
$$\begin{aligned} F &= A\bar{B} + \bar{A}B \\ &= A\bar{B} + A\bar{A} + \bar{A}B + B\bar{B} \\ &= A(\bar{A} + B) + B(\bar{A} + \bar{B}) \\ &= A \cdot \overline{AB} + B \cdot \overline{AB} \\ &= \overline{\overline{A \cdot \overline{AB} \cdot B \cdot \overline{AB}}} \end{aligned}$$



Boolean Algebra



EXAMPLE: Derive the Boolean expression for a two-input Ex-NOR gate to realize with two input NOR gates, without using complemented variables and draw the circuit.



$$\begin{aligned} F &= \overline{A\bar{B} + \bar{A}B} \\ &= \overline{A\bar{B} + B\bar{B} + \bar{A}B + A\bar{A}} \\ &= \overline{\bar{B}(A + B) + \bar{A}(A + B)} \\ &= \overline{\bar{B}(A + B) + \bar{A}(A + B)} \\ &= \overline{B + A + B + A + A + B} \end{aligned}$$

Boolean Algebra



Boolean Functions and their Representations

❖ **Sum-of-Products (SOP) Form:** $f = \bar{A}B + \bar{B}C$

❖ **Product-of-Sums (POS) Form:** $f = (\bar{A} + \bar{B})(B + C)$

❖ **Truth Table Form:** The function is specified by listing all possible combinations of values assumed by the variables and the corresponding values of the function.

Truth Table for $f(A, B, C) = \bar{A}B + \bar{B}C$

Decimal code	A	B	C	f(A, B, C)
0	0	0	0	0
1	0	0	1	1
2	0	1	0	1
3	0	1	1	1
4	1	0	0	0
5	1	0	1	1
6	1	1	0	0
7	1	1	1	0

Standard Sum-of-Product Boolean Functions and their Representations

- ❖ It is also called the Expanded Sum of Products Form or Canonical Sum-of-Products Form.
- ❖ The function is the sum of a number of product terms where each product term contains all the variables of the function either in complemented or uncomplemented form.
- ❖ This can be derived from the truth table by finding the sum of all the terms that correspond to those combinations (rows) for which 'f' assumes the value 1. It can also be obtained from the SOP form algebraically as shown below:

$$\begin{aligned}f(A, B, C) &= \bar{A}\bar{B} + \bar{B}C = \bar{A}\bar{B}(C + \bar{C}) + \bar{B}C(A + \bar{A}) \\&= \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C} + \bar{A}BC + A\bar{B}C\end{aligned}$$

- ❖ A product term which contains all the variables of the function either in complemented or uncomplemented form is called a **minterm**.
- ❖ A minterm assumes the value 1 only for one combination of the variables.
- ❖ An n variable function can have in all 2^n minterms.
- ❖ The sum of the minterms whose value is equal to 1 is the standard sum of products form of the function.
- ❖ The minterms are often denoted as $m_0, m_1, m_2, \dots$, where the suffixes are the decimal codes of the combinations.
- ❖ For a 3-variable function $m_0 = \bar{A}\bar{B}\bar{C}$, $m_1 = \bar{A}\bar{B}C$, $m_2 = \bar{A}B\bar{C}$, $m_3 = \bar{A}BC$, $m_4 = A\bar{B}\bar{C}$, $m_5 = A\bar{B}C$, $m_6 = AB\bar{C}$, and $m_7 = ABC$.

Boolean Algebra



Standard Sum-of-Product Boolean Functions and their Representations

$$\begin{aligned}f(A, B, C) &= \bar{A}B + \bar{B}C = \bar{A}B(C + \bar{C}) + \bar{B}C(A + \bar{A}) \\&= \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}BC + A\bar{B}C\end{aligned}$$

$$f(A, B, C) = m_1 + m_2 + m_3 + m_5$$

$$f(A, B, C) = \sum m(1, 2, 3, 5)$$

Standard Product-of-Sum Boolean Functions and their Representations

- ❖ It is also called **Expanded Product-of-Sums Form** or **Canonical Product-of-Sums Form**.
- ❖ This is derived by **considering the combinations for which $f = 0$** . **Each term is a sum of all the variables.**
- ❖ **A variable appears in uncomplemented form if it has a value of 0 in the combination and appears in complemented form if it has a value of 1 in the combination.**

(the function $f(A, B, C) = (\bar{A} + \bar{B})(A + B)$)

$$f(A, B, C) = (\bar{A} + \bar{B} + C\bar{C})(A + B + C\bar{C}) = (\bar{A} + \bar{B} + C)(\bar{A} + \bar{B} + \bar{C})(A + B + C)(A + B + \bar{C})$$

Standard Product-of-Sum Boolean Functions and their Representations

- ❖ A sum term which contains each of the n variables in either complemented or uncomplemented form is called a **maxterm**.
- ❖ A maxterm assumes the value 0 only for one combination of the variables. For all other combinations it will be 1.
- ❖ There will be at the most 2^n maxterms.
- ❖ The product of maxterms corresponding to the rows for which $f = 0$, is the standard or canonical product of sums form of the function.

$$f(A, B, C) = M_0 \cdot M_4 \cdot M_6 \cdot M_7 \quad \text{or simply as}$$

$$f(A, B, C) = \Pi M(0, 4, 6, 7)$$

The possible minterms and maxterms of a 3-variable function $f(A, B, C)$ are:

$$m_0 = \bar{A}\bar{B}\bar{C},$$

$$m_1 = \bar{A}\bar{B}C,$$

$$m_2 = \bar{A}B\bar{C},$$

$$m_3 = \bar{A}BC,$$

$$m_4 = A\bar{B}\bar{C},$$

$$m_5 = A\bar{B}C,$$

$$m_6 = AB\bar{C},$$

$$m_7 = ABC.$$

$$M_0 = A + B + C,$$

$$M_1 = A + B + \bar{C},$$

$$M_2 = A + \bar{B} + C,$$

$$M_3 = A + \bar{B} + \bar{C},$$

$$M_4 = \bar{A} + B + C,$$

$$M_5 = \bar{A} + B + \bar{C},$$

$$M_6 = \bar{A} + \bar{B} + C,$$

$$M_7 = \bar{A} + \bar{B} + \bar{C}.$$

Boolean Algebra



The possible minterms and maxterms of a 4-variable function $f(A,B,C,D)$ are:

$$m_0 = \bar{A}\bar{B}\bar{C}\bar{D}, \quad m_1 = \bar{A}\bar{B}\bar{C}D, \quad m_2 = \bar{A}\bar{B}C\bar{D}, \dots \quad m_{14} = ABCD\bar{D}, \\ m_{15} = ABCD.$$

$$M_0 = A + B + C + D, \quad M_1 = A + B + C + \bar{D}, \quad M_2 = A + B + \bar{C} + D, \dots$$

$$M_{14} = \bar{A} + \bar{B} + \bar{C} + D, \quad M_{15} = \bar{A} + \bar{B} + \bar{C} + \bar{D}.$$

The possible minterms and maxterms of a 5-variable function $f(A, B, C, D, E)$ are:

$$m_0 = \bar{A}\bar{B}\bar{C}\bar{D}\bar{E}, \quad m_1 = \bar{A}\bar{B}\bar{C}\bar{D}E, \quad m_2 = \bar{A}\bar{B}\bar{C}D\bar{E}, \dots \quad m_{30} = ABCD\bar{E}, \\ m_{31} = ABCDE.$$

$$M_0 = A + B + C + D + E,$$

$$M_1 = A + B + C + D + \bar{E},$$

$$M_2 = A + B + C + \bar{D} + E, \dots$$

$$M_{30} = \bar{A} + \bar{B} + \bar{C} + \bar{D} + E,$$

$$M_{31} = \bar{A} + \bar{B} + \bar{C} + \bar{D} + \bar{E}.$$

Boolean Algebra



EXPANSION OF A BOOLEAN EXPRESSION IN SOP FORM TO THE STANDARD SOP FORM

1. Write down all the terms.
2. If one or more variables are missing in any product term, expand that term by multiplying it with the sum of each one of the missing variable and its complement.
3. Drop out the redundant terms

EXPANSION OF A BOOLEAN EXPRESSION IN POS FORM TO STANDARD POS FORM

1. Write down all the terms.
2. If one or more variables are missing in any sum term, expand that term by adding the products of each of the missing variable and its complement.
3. Drop out the redundant terms.

Example:

Expand $\bar{A} + \bar{B}$ to minterms and maxterms.

$$\begin{aligned}\bar{A} + \bar{B} &= \bar{A}(B + \bar{B}) + \bar{B}(A + \bar{A}) \\ &= \bar{A}B + \bar{A}\bar{B} + \bar{B}A + \bar{B}\bar{A} \\ &= \bar{A}B + \bar{A}\bar{B} + A\bar{B} + \bar{A}\bar{B} \\ &= \bar{A}B + \bar{A}\bar{B} + A\bar{B} \\ &= 01 + 00 + 10 \\ &= m_1 + m_0 + m_2 \\ &= \Sigma m(0, 1, 2)\end{aligned}$$

Boolean Algebra



Example:

Expand $A + B\bar{C} + AB\bar{D} + ABCD$ to minterms and maxterms.

$$\begin{aligned} A &= A(B + \bar{B})(C + \bar{C})(D + \bar{D}) \\ &= ABCD + ABC\bar{D} + AB\bar{C}D + AB\bar{C}\bar{D} + A\bar{B}CD + A\bar{B}C\bar{D} + A\bar{B}\bar{C}D + \\ &\quad A\bar{B}\bar{C}\bar{D} \end{aligned}$$

$$B\bar{C} = B\bar{C}(A + \bar{A})(D + \bar{D}) = AB\bar{C}D + AB\bar{C}\bar{D} + \bar{A}B\bar{C}D + \bar{A}B\bar{C}\bar{D}$$

$$AB\bar{D} = AB\bar{D}(C + \bar{C}) = ABC\bar{D} + AB\bar{C}\bar{D}$$

$$\begin{aligned} \text{or } A + B\bar{C} + AB\bar{D} + ABCD &= ABCD + ABC\bar{D} + AB\bar{C}D + AB\bar{C}\bar{D} + A\bar{B}CD + A\bar{B}C\bar{D} \\ &\quad + A\bar{B}\bar{C}D + A\bar{B}\bar{C}\bar{D} + \bar{A}B\bar{C}D + \bar{A}B\bar{C}\bar{D} \\ &= m_{15} + m_{14} + m_{13} + m_{12} + m_{11} + m_{10} + m_9 + m_8 + m_5 + m_4 \\ &= \Sigma m(4, 5, 8, 9, 10, 11, 12, 13, 14, 15) \end{aligned}$$

In the SOP form, the minterms 0, 1, 2, 3, 6, and 7 are missing. So in the POS form, the maxterms 0, 1, 2, 3, 6, and 7 will be present. Therefore, the POS form is

$$\Pi M(0, 1, 2, 3, 6, 7)$$